2026

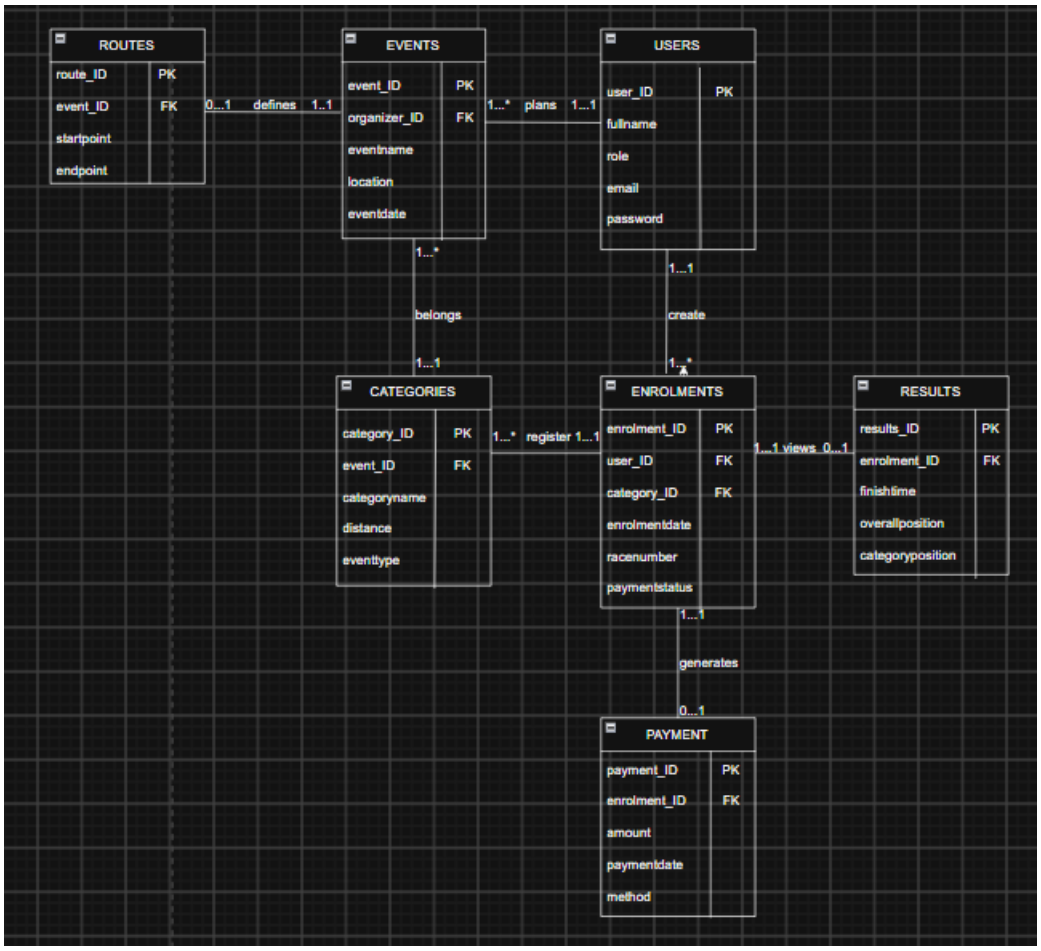
PROGRAMMING 6212

PART1

SIBONGUMUSA MADI

Section A Entity relationship diagram (ERD)

Created using : [Untitled Diagram - draw.io]



Section B- API endpoint plan

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user account as either an Organiser or Participant of the race.	None (Public)	{ "name": "String", "email": "String", "password": "String", "role": "Organiser | Participant" }	**200 Created** - User object and JWT token. **400 Bad Request** - Invalid fields. **409 Conflict** - Email already exists.
POST	/api/auth/login	Authenticates an existing user and issues an access token.	None (Public)	{ "email": "String", "password": "String" }	**200 OK** - User profile and JWT token. **401 Unauthorized** - Invalid credentials.
GET	/api/users/profile	Fetches current authenticated user's profile details.	Any (Logged in)	None	**200 OK** - User profile details. **401 Unauthorized** - Missing token.
PUT	/api/users/profile	Fetches current authenticated user's profile details.	Any (Logged in)	{ "name": "String", "email": "String" }	**200 OK** - Updated user profile. **400 Bad Request** - Invalid data.

					401 Unauthorized - Token invalid.
GET	/api/events	Retrieves list of all events, filterable by event type (run, walk, cycle).	Noun (Public)	None	**200 OK** - Array of event objects.
GET	/api/events/{id}	Retrieves detailed information for a specific event.	None (Public)	None	**200 OK** - Event details. **404 Not Found** - Event ID does not exist.
POST	/api/events	Creates a new road running, walking, or cycling event.	Organiser	{ "name": "String", "description": "String", "date": "DateTime", "location": "String", "distance": "Number", "eventType": "run | walk | cycle" }	**200 Created** - Created event object. **400 Bad Request** - Validation failure. **403 Forbidden** - Role mismatch.
PUT	/api/events/{id}	Updates details of an existing event managed by the organiser.	Organiser	{ "name": "String", "description": "String", "date": "DateTime", "location": "String", "distance": "Number", "eventType": "run | walk | cycle" }	**200 OK** - Updated event object. **403 Forbidden** - Not event owner. **404 Not Found** - Event not found.
DELET	/api/events/{id}	Deletes an event from the system.	Organiser	None	**200 OK** - Success message.

					403 Forbidden - Not event owner. **404 Not Found** - Event not found.
GET	/api/events/{eventId}/categories	Retrieves all age/distance categories associated with a specific event.	Non (Public)	None	**200 OK** - List of categories. **404 Not Found** - Event does not exist.
POST	/api/events/{eventId}/categories	Creates a new category (e.g., Senior, 21km) for an event.	Organiser	{ "categoryName": "String", "description": "String" }	**201 Created** - Created category record. **400 Bad Request** - Invalid data. **403 Forbidden** - Not event owner.
POST	/api/enrolments	Enrols the logged-in participant into an event category.	Participant	{ "eventId": "Integer", "categoryId": "Integer" }	**201 Created** - Enrolment record. **400 Bad Request** - Missing parameters. **409 Conflict** - Already enrolled in event.

GET	/api/events/{eventId}/enrolments	Retrieves all participant enrolments for a specific event.	Organiser	None	**200 OK** - List of enrolled participants. **403 Forbidden** - Not event owner. **404 Not Found** - Event not found.
GET	/api/users/enrolments	Retrieves all event enrolments for the currently logged-in participant.	Participant	None	**200 OK** - List of participant's active enrolments.
POST	/api/events/{eventId}/results	Captures finish time and position for a participant after an event.	Organiser	{ "participantId": "Integer", "finishTime": "String", "position": "Integer" }	**201 Created** - Created result record. **400 Bad Request** - Invalid fields. **404 Not Found** - Participant/Event not found.
GET	/api/events/{eventId}/results	Retrieves overall finishing results for a given event.	None (Public)	None	**200 OK** - List of event finishing results. **404 Not Found** - Event not found.
GET	/api/users/results	Retrieves complete historical race performance records for	Participant	None	**200 OK** - Historical performance list.

		the logged-in participant.			**401 Unauthorized** - Invalid session.

Section C SQL: results

Created using: sql server management studio

